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1.1

Introduction to HCI

- In 1992, psychologist at Apple studied how people organized the rapid flow of info in their work spaces
- They observed people pile of materials, then they design system that mimic that ability
- They had better understanding on user
- HCI helps us understand users better
- HCI is about research
    - Needfinding
    - prototyping
    - evaluation
- HCI also is about Design

    - distributed cognition
    - mental models
    - universal design

AI and HCI

- Marty Hurst, made the observation, as technologies improves, computer interaction moves toward being more natural
- Folders, mimics files in cabinet
- Touch screens and voice recognition, touch them like we touch things in the physical world
- ChatGPT interact with us, in a very human way
- Jakob Nielsen, AI is introducing the 3rd user-interface paradigm in computing history, in which human tell computers what they want, now how to do it

1.2

1.3

Introduction to exploring HCI

- 3 areas,
    - technology: emerging technological capabiliities
    - domain: pre-existing areas that could be significantly disrupted by computer interfaces like healthcare, education, entertainment
    - ideas: theories how people interact with interfaces, and world around them

Tech: VR

- Interaction and visualisation
- Usually video games, tourism, commerce, education, healthcare
- VR therapy for trauma is new

Tech: AR

- Incredible opportunities enhancing realities
- Automated speech translation
- Tours in museums
- Restaurant review or menu
- What putting cameras would mean? to check in politics topic

Tech: UbiComp and wearables

- Figuring out how to build smart device that takes away cognitive load from users
- Rise of wearables like smart watches, fitness trackers
- Advance hearing aids in prostethic limbs

Tech: Robotics

- Physical construction abilties and artificial intelligence
- Human-robot interaction: how people interact with robots
- How do we ensure that robots don't harm humans through faulty reasoning
- How to integrate robot in social life
- How to deal loss of demand for human work
- Voice and touch? Input properly received and understood?

Tech: Mobile

- Support experiences from navigation to stargazing
- Presents some fascinating challenges to address
    - can't do all works that can be done on computers, like writing essays

Idea: Context-sensitive computing

- Context is a fundamental part of human interactions
- Give computer interfaces the contextual knowledge that humans have in their everyday lives
- E,g, use mobile phone differently at sidewalk, sofa or sidewalk

Idea: Gesture-based interaction

- Hand gestures, big, small, narrow, wide
- Microsoft Kinect, healthcare to gaming
- Wristband that reacts to certain motions
- Maybe it;s possible to type in the air with finger one day

Idea: Pen- and touch-based interaction

- Shorten the distance between user and task we want to accomplish
- Mouse and screen, look at screen and move mouse
- Touch screen, directly touch what we want to interact with

Idea: Information visualization

- Conclusion are only useful if we can understand them
- Computers give a powerful way to represent data in complex, animated and interactive ways
- match user's mental model with the task at hand
- A visualization is just a representation of data

Idea: CSCW

- Computer supported cooperative work
- How do we use computers to support people working together?
- Distributed teams are an example of CSCW

Idea: Social Computing

- How computers affect how we interact and socialize
- Chat online, emojis? emoticons?

Domain: Special needs

- help people with special needs
- Robotic prosthetics
- How the person intends to use the limbs to perform tasks

- How do we communicate data to blind person?
- VR for people with paralysis
- Autonomous wheelchair

Domain: Education

- Designing educational technology is part of HCI
- Cognitive load of students must be on materials itself, not the interface itself

Domain: Healthcare

- Visualize for better diagnosis
- Certain decision on patient's wellbeing
- Fitbit for fitness modelling
- Virtual reality for therapy

Domain: Security

- It has to be usable, if not people wont use it
- HCI can improve usability of security in number of ways
- CAPTCHAs, now as simple as checkbox
- Less frustating to users
- Password with complexities, rank like a game
- Incentivize users to create strong passwords

Domain: Games

- A good game has good logical mapping between actions and effects
- Should feel like user are interacting in the game world
- Many of criticsm are because of HCI, penalty of failures are too high/ too low!

Reflections: Exploring HCI

- Apply what we learn in our chosen interated area

Conclusion

- Select a topic that we find interested in, read more about it, and reflect on how HCI concepts apply to it

2.1

Introduction to Principles

- Focus on task in HCI, not on tools and interfaces on their own
- Talk about role of interface and how it mediates user and task
- Different view on roles on the system
- UX at multiple levels

Interfaces: Between Users and Tasks

- Our focus should be user and task, not interface
- To design good interface, need to find user's goal and what they want to achieve
- In thermostat example, by focussing ont he task instead of just the interface, we can come up with more revolutionary design like Nest, instead of the traditioanal thermostat

Tips: identifying the task

- Video of someone paying by swiping her credit card. What is the task? Swiping is too narrow so answer is paying for groceries
- 5 quick tips
    - Watch real users
    - Talk to them - what their goals and motives
    - Start small - individual interactions
    - Abstract up - working from smaller obs, try to understand task they're trying to complete
    - You are not your user - leave behind previous experience, biases

Usefulness and Usability

- Useful: interface allows the user to achieve their goal, but it is the lowest bar,
- Usability: most important, how easy it is to use the interface to achieve their goal, Google Maps to replace paper map

Views of the User: Processor

- Need to understand the role human has in the system
- Processor, predictor, participant
- Interface must fit within human limits
- Evaluated by: quantitative experiments, take numeric measurements of how fast they can complete a task and etc

Views of the User: Predictor

- Care deeply about knowledge, experience and thought process
- Want them to be able to get input and give output
- Interface must fit with knowledge
- Evaluated by qualitative studies: ex situ studies, cognitive walk thru to understand how user thinks about the system
- We still focus on one user, and one task, but we might wanna see it more broadly. Hence participant view

Views of the User: Participant

- Not just interested in their head, but around them. like what kind of people they interacting with
- What's competing for their attention, their cognitive load
- Interface must: fit with the context: actually interact with the system where they are need
- Evaluated by: in situ studies (within the authentic context), context that are relevant

Views of the User and Psychological Schools of Thoughts

- Processor - behaviorism: aim to provide systematic way of investigating behavior in human and animals

- Little Robert experiment
- Attempts to understand behavior, just by observing the behavior
- Our design process focus on testing observable behaviors
- Predictor - cognitivism: concern with what's going on inside the mind
- Memory creativy and etc
- Care about what the user is thinking
- Predictive model of the user, what the user is predicting
- We ask question, what do they predict the ourcome of the action to be? is it the right action?
- We want to get inside user's head
- Participant - Functionalism, System: Context in broader environment
- Look at the interaction of both above at the larger system
- We look beyond just the user and the interface

Designing with three views

- Redesigning Tesla infotainment system

Processor

- Pros: May use existing data
- Pros: Enables objective comparisons
- Cons: Doesnt find reason for differences
- Cons: Cant differentiate by expertise
- Cons: Help optimize, not redesign

Predictor

- Pros: More complete picture of interaction
- Pros: Targets different levels of expertise
- Cons: Analysis may be expensive, plain text of interviews, surveys (need a lot of effort)
- Cons: Analysis subject to biases

- Cons: Ignore broader Interaction context (test in labs, vs real world)

Participant

- Pros: Evaluates interaction in real world context
- Pros: Captures authentic user attention
- Cons: Expensive to perform and analyze
- Cons: Requires real, functional interfaces
- Cons: Subject to uncontrollable variables

	Processor	Predictor	Participant
Requirement	Fit within human limits	Fit within user's knowledge	Fit within usage context
Evaluation	Quantitative experiments	Qualitative studies	*In situ* (real-world) studies
Pros	Objective comparisons May use existing data	Fuller picture May target novices	Authentic context Faithful user attention
Cons	Expert-focused What, not why Optimize, not redesign	Expensive to analyze Prone to biases Ignores context	Expensive to perform Needs real interfaces Subject to interference

User experience: Sans Design

- UX goes beyond task via interface
- Based on age, sex, culture, prior experience and etc
- Examines emotional response, aesthetics, values and etc
- 3 levels; Individual, Group, and Society
- Make sure user feels represented by the system

Design Challenge: Morgan

Morgan loves audiobooks, takes notes and bookmarks and stuff

- As a processor, we might simply look at what information is communicated to Morgan, when, and how.

- As a predictor, we might look instead at how the interface meshes with Morgan's needs with regard to this task, how easy it is to access, how easy the commands are to perform, and so on.
- As a participant, we might look at the broader interactions between this interface and Morgan's other tasks and social activities.

2.2

Feedback cycles are fundamentals

- Intelligence: ability that must be gain thru feedback cycles

Gulf of Execution

- Two general challanges:
  - Users's interaction with the task thru the interface: gulf of execution
    - How do i know what i can do
  - Task's return to the user of the output via the interface: gulf of evaluation
- First: Identify intention
- Second: Identify actions
- Third: Execute actions
- GoEx: Takes the user from understand their own goal to understand goal in the context of the system to understand the actions necessary to realize the goal to executing those actions

5 tips of GofEx

- Make functions discoverable
  - How they know what they can do? documentation? discoverablke
- Let the user mess around
  - Avoid any button that can cause irreversible damage
- Be consistent with other tools
  - Familiarity
- Know your user
  - Identify intention, identify actions and execute actions
- Feedforward
  - Feedback about what user might want to do

    - Information on what would happen if keep doing what we;re doing

Gulf of Evaluation

- The output of the actions that the user took
- Interface output -> Interpretation -> Evaluation
- On-demand video shows black, not showing buffering icon to help user understand what's going on, evaluate the situation

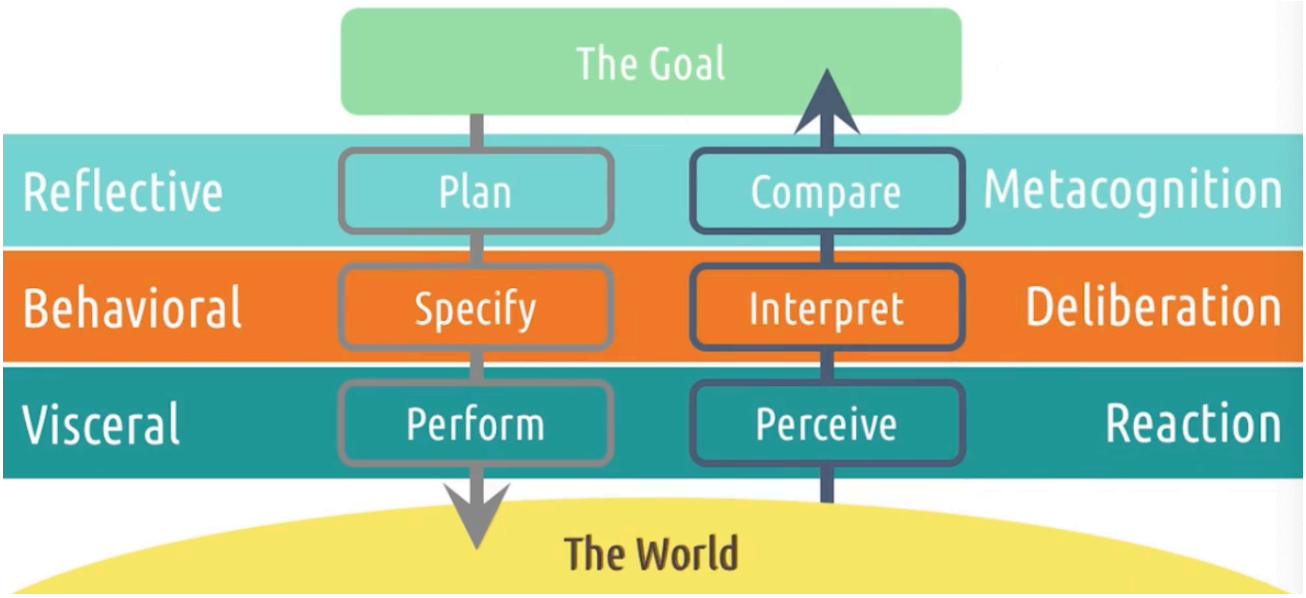
Example: Thermostat

- Feeling the heat is good, hearing the heater turn on might not
- Mark the thermostat the heat is on to lower the gulf of evaluation

5 tips of GofEv

- Give feedback constantly
    - Dont wait, give them feedback that the input was received, and what input is received
- Give feedback immediately
    - Tap on app, icon briefly grays out when opening app if there's buffer
- Match feedback to the action
    - Subtle action should result in subtle feedback
- Vary your feedback
    - Think about how auditory or haptic feedback
- Leverage direct manipulation
    - Dragging stuff around, making things large

Norman's Feedback Cycle Stage



Norman ask 7 question when designing these interfaces

- How easily can one
  - determine the function of the device
  - tell what actions are possible
  - determine the mapping from intent to movement
  - actualy perform the physical movement
  - tell what state the system is in
  - tell if the system is in the desired state
  - determine the mapping from state to interpretation

Conclusion

- Feedback cycles: exchange of input and output between user and system to share some goal
- GoFEx: The distance between making some change in the system and evaluate whether or not the goal was accomplished
- 7 questions to bridge those goals
- Understanding method for crossing the goals

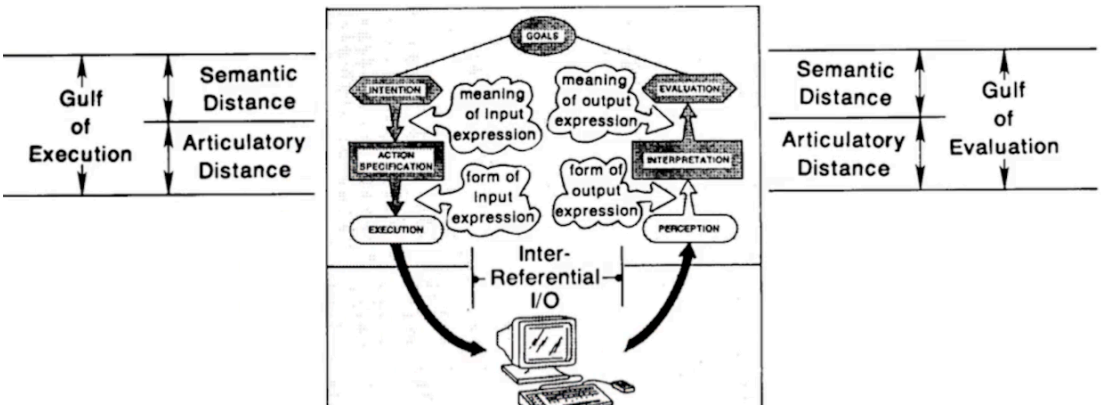
2.3

Direct Manipulation The Desktop Metaphor 1

- Files and folders in desktops are to mimic physical files and folders
- Pick files and put them in folders
- Would it narrow the gulf of execution if we mimic physical actions?
- Previously use of command line interfaces to manipulate files
- Then computer mouse came along, made it easier to point and click
- GUI made it easier to drag and drop files into folders
- However, there is still a gap for novice users
- Touch screens make it even easier to directly manipulate objects on screen

Paper Spotlight: Direct Manipulation Interfaces

- Hutchins, Hollan, and Norman (1985)
- Becoming a more common design strategy
- Two aspect of directness
  - Distance between the user's goal and the system. The greateer the cognitive load, the less direct the interface
    - Semantic distance: user's goal and expression in the system, how hard is it to do
    - Articulatory distance: expression and execution, how hard is to do the action when you know what to do
  - Direct engagement
    - If we're moving files, we want to see the files moving
    - In game, we should directly control our avatar
    - Direct manipulation is a powerful tool to shorten the distance



Exploring HCI: Direct Manipulation and VR

- Able to view what they deign in 3d
- Move around the object to get angles
- Directly manipulate the object with hands

Babies and Direct Manipulation

- Watching a baby to user interface
- Games made for cats, even better!

Making Indirect Manipulation More Direct

- Mac touchpad: pinching to zoom in and out, swipe to bring up notification center
- Spread out fingers to clear the windows, like clearing off your desk, but isnt really direct, but the motion mimics physical action

Exploring HCI: Direct Manipulation

- Direct Manipulation is contingent on the immediate feedback that maps directly to the interaction
- How does VR do direct manipulation kinesthetically?

Invincible Interfaces

- We make the interface to be invisible
- The feel like they are not using an interface, but working directly with the task
- Draw with an apple pen on iPad, it feels like drawing on paper
- Contrast that with drawing on a computer with a mouse, it feels indirect, ugly writing

Good Design - Bad Design Invisible Interfaces

- Good design: interfaces that are metaphorically invisible
- Bad design: interfaces that are literally invisible, kind of xD, must give direct feedback else users get lost without any clue of what is going on

Invisibility by Learning

- Driving, discover i am headed for someone, we hit the brake and rotate the steering wheel
- We practice these actions so much that we don't think about them anymore
- Driving is important enough that we can have a higher learning curve
- But interface we design, we can't expect users to go through a high learning curve
- Our goal is below

Invisibility by Design

- Should feel immediately as if they're interacting with the task, underlying the interface
- When we do things right, people won't be aware we done anything at all

5 tips: Invisible Interfaces

- Use affordances
  - visual design of the interface is just how it's supposed to be used
  - Buttons for turning, dials for turning
- Know your user
- Differentiate your user
  - Novice vs expert
  - copy and past in menu vs cntrl-c, ctrl-v
- Let your interface teach
  - When user select copy from Edit menu, they see the hotkey for shotcur
- Ask the user
  - How the user feels
  - Note especially they're talking about the task or the interface
  - If the talk about the interface, then it's pretty visible

Reflections: Invisible By Design

- usually in game, people move joystick up to look up
- but some people move joystick down to look up, like in airplane
- in portal 2 game, instead of updating in control settings menu, they do it in-game
- If buzzer sounds, user immediately lookup, then the control is updated

Design Challenge: Universal Remote

- Voice interface for remote control
  - Command is non-discoverable, need it to understand natural language
  -

2.4

Information Processing

- Focus how a person make sense of input, don't care about what the action of the world is

Sensation and Perception: Visual

- Morgan has eyes - color and movement
- As a woman, less likely to be color blind
- Peripheral vision is more sensitive to motion, less to color and detail
- As Morgan gets older, less able to see in low light, need to design for that

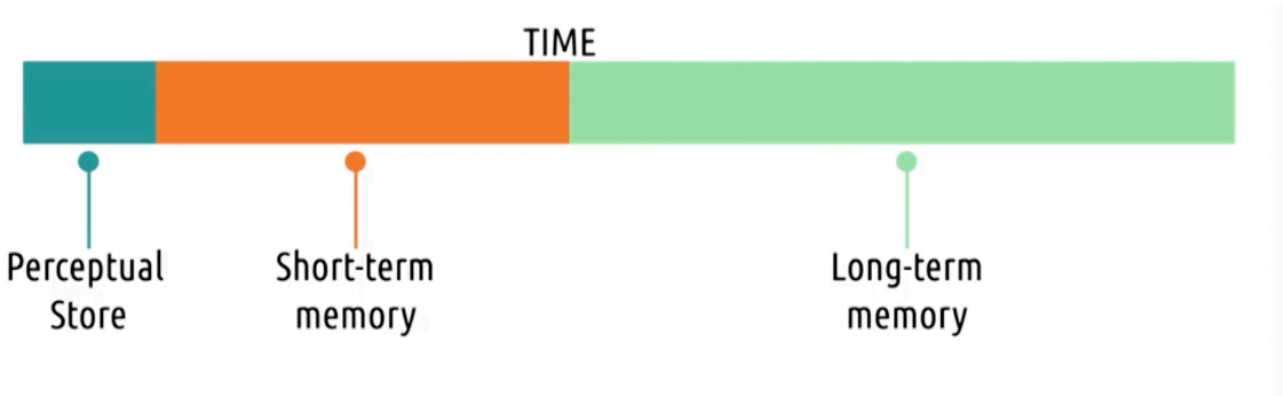
Sensation and Perception: Auditory

- Morgan has ears - pitch and loudness
- Difference between far or near sound
- Morgan can't close ear or move ear direction, need to design for that

Sensation and Perception: Haptic

- Skin can feel things
- Pressure, vibration and temperature
- Can't filter out touch feedback, but unlike listening, it is only available to the person touching

Memory



Memory: perceptual store

- Working memory, very short term, few seconds
- Three parts:
  - Visuospatial sketchpad: visual info for active manipulation, picture in head
  - Phonological loop: verbal or auditory info, store the sound you heard recently
  - Episodic buffer: integrate info across systems to put things in place
- All are coordinated by central executive,
- Expertise or rehearsal delays the decay of the perceptual buffer

Memory: Short-term and Chunking

- Avoid requiring users to remember too much at once
- More recent research, only four to five chunks at a time
- 2 principles
  - Chunking - grouping together several items into a single meaningful item e.g.
    - Right most the easiest to remember, we dont remember letters, we remember words
    - Middle next, then left most the hardest
    - Recognizing is easier than recall!

Memory: Short-term takeaways

- We don't want user to remember too much at once
- Four to five chunks at a time
- When, possible, we should leverage recognition over recall

Memory: Long term memory

- Unlimited store of memory, but harder to put
- Need to put in short term memory several times to get into long term memory
- E.x. remembering flash cards, formulas, vocab

Cognition

Cognition: Learning

- Procedural learning: how to do something
  - Doing work on computer, sports, something that we do
- Declarative learning: knowledge about something
  - What we know in head and know what to answer when asked
  - E.x. what's hotkey for copy
- In all likelihood, when asked about what's the hotkey for paste, we mentally simulate doing it
- Procedural knowledge is more about what we do at HCI
- When we strong PK, we unconsciously competent to do things
- Declarative knowledge is how we communicate with others about what we do

Cognition: Cognitive Load

- Brain like CPU, take resources to do things

- Has two working applications
  - Reduce cognitive load on interface
  - want to understand the context on what's going on when user use the interface, need to understand their cognitive load

5 tips: Reduce Cognitive Load

- Use multiple modalities
  - Describe things verbally, and also present them visually
- Let the modalities complement each other
  - focus on letting each modality support, illustrate the other instead of competing
- Give user control of the pace
  - Menus disappearing, selections needed to be made, bla bla
- Emphasize essential content and minimize clutter
  - Design interface like common actions
- Offload task
  - Pay closely what user has to do or remember at every stage
  - See where we can offload
  - Show what they entered in previous screen and etc

Motor System

- Error in what we physically able to perform at a given time
- E.g. Small close button, user accidentally tap on other button, especially in different context like running while using phone
- Need to make our interface more tolerant to errors if situation is not forgiving
- E.g. double tap to close the app when in lock mode menu, or make important buttons larger (might not possible in mobile screen)

2.5

Design Principles & Heuristics

- 6 principles from Don Norman's The Design of Everyday Things
  - Discoverability

  - Simplicity
  - Affordances
  - Mapping
  - Consistency
  - Feedback

The Sets

- 4 sets
  - Don Norman's 6 principles, recent one 7
  - Jakob Nielsen's 10 ten design heuristics
  - Larry Constantine's and Lucy Lockwood's 6 principles
  - Ronald Mace's 7 principles of universal design
- In our case, we learn about 15

	Don Norman	Jakob Nielsen	Constantine & Lockwood	Universal Design
Discoverability	✓	✓	✓	
Simplicity		✓	✓	✓
Affordances	✓			
Mapping	✓	✓		
Perceptibility	✓*	✓		✓
Consistency	✓	✓	✓	
Flexibility		✓		✓
Equity				✓
Ease				✓
Comfort				✓
Structure			✓	
Constraints	✓	✓		
Tolerance		✓	✓	✓
Feedback	✓	✓	✓	
Documentation		✓		

Discoverability

- Norman: Is it possible to even figure out what actions are possible and how to perform them
- Nielsen: Minimize the user's memory load by making objects, actions, and options visible... instructions for use of the system should be visible or easily retrievable

whenever appropriate
- Constantine: The design should make all needed options and materials for a given task visible without distracting the user with extraneous or redundant information
- Should be visible
- E.g. powerpoint toolbars,
- Not visible - ctrl + shift + 4 to screenshot in macOS
- Walk on the line between discoverability and simplicity

Simplicity

- More discoverable, more complex
- Nielsen: Dialogues should not contain information which is irrelevant or rarely needed. Every extra unit of information in a dialogue competes with the relevant units of information and diminishes their relative visibility.
- Constantine: The design should make simple, common tasks easy to perform
- Mace: use of the design is easy to understand, regardless of the user's experience, knowledge, language skills, or current concentration level
- E.g. BSOD in Windows,
- Nielsen: the user should only see what they need at the moment

Affordances

- Both simple and usable
- Physical design of an object should suggest how it is to be used
- E.g. buttons are for pressing, dials are for turning
- Norman: An affordance is a relationship between the properties of an object and the capabilities of the agent that determine just how the object could possibly be used. The presence of an affordance is jointly determined by the qualities of the object and the abilities of the user.

Affordance Vocabulary

- Affordance: inherent property of a device, we can't add affordance to an object, we use signifiers to communicate affordance
- Perceived Affordance: property attributed ot the object by the user
- Signifiers: in context instruction like arrow where to swipe, create affordance, match perceived affordance to actual affordance

Mapping

- Norman: mapping is a technical term, meaning the relationship between the elements of two sets of things
- Nielsen: The system should speak the user's language, with words, phrases and concepts familiar to the user, rather than system-oriented terms. Follow real-world conventions, making information appear in a natural and logical order.
- Mapping vs Affordances
  - Affordance: about creating interfaces where their designs suggest how they are to be used, horizontal bar of color slider
  - Mapping: creating interfaces where the design makes it clear what the effect will be when using them

Design Challenge: Mapping and Switches

- E.g. light switches -
  - small letter for each switch
  - order the switches in the same order as the lights in the room

Perceptibility

- User's ability to perceive the state of the system
- Nielsen: The system should always keep users informed about what is going on, through appropriate feedback within reasonable time.
- Mace: The design communicates necessary information effectively to the user, regardless of ambient conditions or the user's sensory abilities.
- Norman: Feedback must be immediate and informative, poor feedback can be worse than no feedback at all.
- Fan with lights with rope switch
  - One pull for light, two pulls for fan
  - But no feedback on which state it's in
  - Solution: different pull lengths for light and fan, but still no mapping

Consistency

- Norman: Consistency in design is virtuous. It means that lessons learned with one system transfer readily to others

- Nielsen: Users should not have to wonder whether different words, situations, or actions mean the same thing. Follow platform conventions.
- Constantine: The design should reuse internal and external components and behaviors, maintaining consistency with purpose rather than merely arbitrary consistency.
- Consistent within and across the interface
- Links we use in text on website, always blue and underlined

Consistency: The curious case of ctrl-Y

- In most Windows applications, ctrl-Y is redo
- In Visual Studio, ctrl-Y is delete line?? Redo is something else
- Because in Visual Basics, ctrl-Y is delete line. Microsoft choose to be consistent with VB, but violate the Windows standard

Flexibility

- Nielsen: Accelerators, may often speed up the interaction for the expert user such that the system can cater to both inexperienced and experienced users. Allow users to tailor frequent actions.
- Mace: The design accommodates a wide range of individual preferences and abilities.
- Not forcing them into one way of doing things, but allow them to customize

Equity

- Mace: The design should accommodates a wide range of individual preferences and abilities.
- Mace: The design is useful and marketable to people with diverse abilities.
  - Provide the same means of use for all users: identical whenever possible; equivalent when not.
  - Avoid segregating or stigmatizing any users.
- E.g. requirements for password reset
  - Novices might not know level of security needed as experts
  - Give tips on how to make strong password

Ease and Comfort

- Mace: The design can be used efficiently and comfortably and with a minimum of fatigue.
- Mace: Appropriate size and space is provided for approach, reach, manipulation, and use regardless of user's body size, posture, or mobility.

Structure

- Overall architecture of the interface
- Constantine: Design should organize the user interface purposefully, in meaningful and useful ways based on clear, consistent models that are apparent and recognizable to users.

Constraints

- A big part of usability is accounting for user errors
- Manya agrees that there's no such thing as user error. If the user makes an error, it was because the system was not structured in a way to prevent or recover from the error
- Avoid: prevent user from making error
- Norman: Constraints are powerful clues, limiting the possible actions.
- Nielsen: Even better than good error messages is a careful design which prevents a problem from occurring in the first place. Either eliminate error-prone conditions or check for them and present users with a confirmation option before they commit to the action.
- E.g. password reset, it shows constraints, and shows we violate them
- Norman 4 types of constraints
    - Physical: physical properties limit possible actions, e.g. shape of plug
    - Cultural: rely on social conventions to control possible actions, e.g. red means stop
    - Semantic: rely on knowledge of the world to control possible actions, e.g. car won't start if not in park

    - Logical: rely on reasoning about the situation to control possible actions, e.g. three buttons but only two can be pressed at a time

Reflections: Constraints

- Constraint is active in user's way, so it's obvious
- E.g. car options creen
    - Can only change when car is not moving
    - Introduce constraint to prevent user from changing settings when driving

Tolerance

- Two principles of how we deal with errors
- Tolerance means users shouldn't be at risk of causing too much trouble accidentally
- Nielsen: Users often choose system functions by mistake and will need a clearly marked "emergency exit" to leave the unwanted state without having to go through an extended dialogue. Support undo and redo.
- Lockwood: The design should be flexible and tolerant, reducing the cost of mistakes by allowing undoing, redoing while also preventing errors wherever possible
- Mace: The design minimizes hazards and the adverse consequences of accidental or unintended actions.
- Jef Raskin: A computer shall not harm your work or, through inaction, allow your work to come to harm.

Feedback

- System should give feedback to user about what is going on
- Norman: Feedback must be immediate and informative, poor feedback can be worse than no feedback at all
- Nielsen: Error messages should be expressed in plain language (no codes), precisely indicate the problem, and constructively suggest a solution.
- Constantine: The design should keep users informed of actions or interpretations, changes of state, and errors or exceptions

Documentation

- One goal of usable design is to avoid documentation

- Nielsen: Even though it is better if the system can be used without documentation, it may be necessary to provide help and documentation. Any such information should be easy to search, focused on the user's task, list concrete steps to be carried out, and not be too large.

Exploring HCI: Design Principles and Heuristics

- If we work in complex system, we need to define what simplicity means in that context
- If we design something that people with disability can't use, we risk big problems both ethically and legally

Other sets or principles

- 29, but shortened to 15 above
- Conceptual models by Norman
- Dix and Finlay's
  - learnability
  - flexibility
  - robustness
- Raskin's Human Interface
- Foley's Computer Graphic
- Designing Effective Speech Interfaces by Cohen et al.

3.1

Introduction to Methods

- We dont want to build novel interface just for the sake of novelty
- WE want novelty to have a purpose, need to have strong understand user's task

User-centered Design

- Design that considers user through out the design process
- Prioritizing user;s need while also recognizing that we dont know the user's need

- Steps
  - examine user;s need in depth
  - present to user to get feedback
  - evaluate the design with real users
  - we have to interact with users to get feedback

Principles of User-centered Design

- Design is based upon explicit understanding of users, task and environment
- Users are unvolved throughout the design and development
- Driven and refined by user-centered evaluation
- The process is reiteraive (constant improvement)
- The design addresses the whole user experience
- THe design team includes multidisciplinary skills and perspective

Stakeholders

- Primary: user, who use our tool directly
- Secondary: Dont use the tool directly, but interact with the output in some way
- Tertiary: Never interact with tool or output, impacted by existence of tool

e.g. new grade book tool

- Primary: teachers
- Secondary: Parents (interact with the output)
- Tertiary: Students (affected by the system)
- In design this tool, we need to remember of all three
- Avoid over-controlling parents, impact of design to all stakeholders

The Design Life Cycle

- Four phase design life cycle
  - Needfinding: gather task user tries to perform
  - Design Alternatives: early ideas
  - Prototyping: Take the highest potential to take to the user

    - Evaluation: user eval and feedback
- Each time we go thru the cycle, our idea and prototypes improve

Methods for the Design Life Cycle

- Tools in our toolbox
- Goal of this class is not just to understand more HCI, but also to understand how big the field of HCI actually is

AI in the Design life cycle

- Three primary ways
    - AI can be used as a jump start for user perspective (ask their needs).
        - Negative side, might constraint our thinking too early. Too narrow to what AI suggested
    - AI can be a great tool to practice with as if it's your user
        - Rubber duck debugging
        - Wouldnt put alot of stake to what it says, but lets us practice what we ask and respond accordingly
    - AI tools that help accomplish certain parts of design life cycle faster
        - paper prototyping
- At no point during any of this, did he ever suggest using AI instead of interacting with users
- Info from AI is no substitute from user
- Can get early base line
- Can get practive question
- Create prototypes

Design life Cycles Meet Feedback Cycles

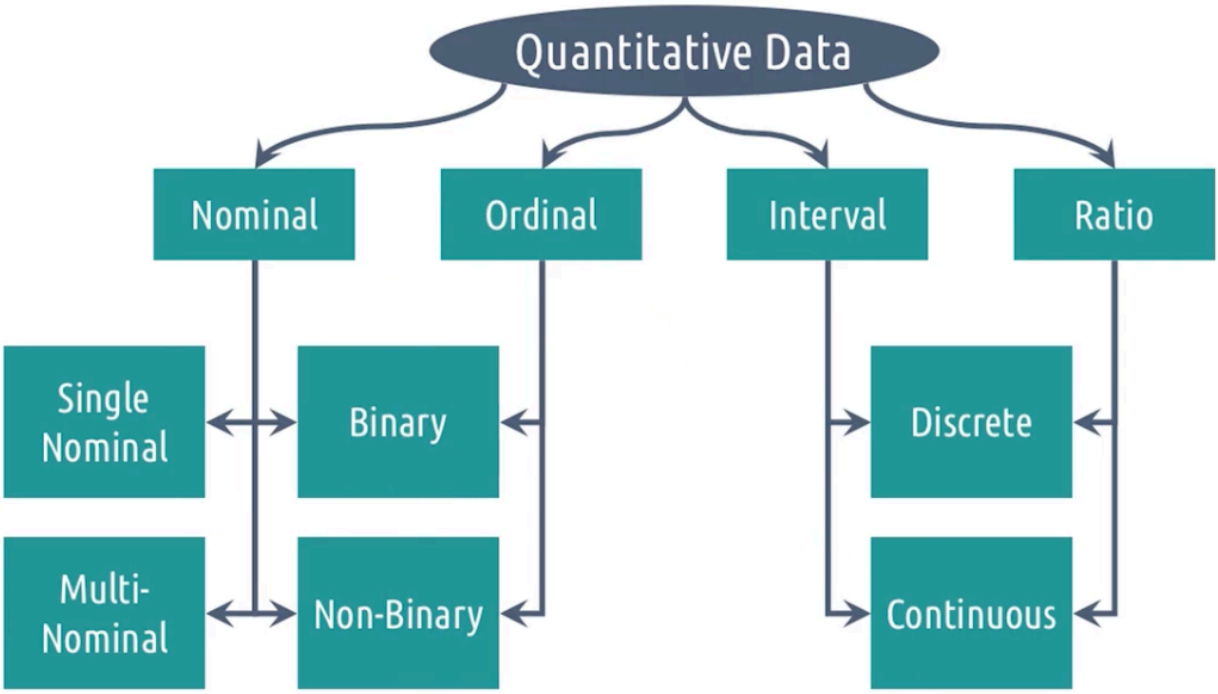
- In a feedback cycle, the user does something in the interface to accomplish some goal. And then judges based on the output from the interface, whether the goal was accomplished, then they repeat and continue.
- In HCI, we're brainstorming and designing interfaces to accomplish goals. And then based on the output of our evaluations, we judge whether or not the goals of the interface were actually accomplished, then we repeat and continue.
- In many ways, we're doing the same things that our users are doing, trying to understand how to accomplish a task in an interface.

- Only in our case, our interface is the tools to build and evaluate interfaces and our goal is to help them accomplish their goal.

Qualitative vs Quantitative Data

- Qualitative
  - Pretty much everything else, observations described or summarized non-numerically
  - Descriptions, accounts, observations, open-ended survey, bug reports
  - Broader picture of what;s happening
  - More prone to biases
  - Can convert it to quantitative: count the number of surveys
  - Provides the how or why
- Quantitative
  - Anything numeric, observations described or summarized numerically
  - Formal tests, summaries
  - Narrow view of what we are interested in examining
  - Provides the what
- They both difference purpose
- Mixed method: mixture of qualitative and quantitative data to paint a more complete picture of the results

Types of Quantitative Data



- Nominal
  - Categorical data
  - e.g. How do you typically commute to work?
  - Single Nominal
    - One option from question above
  - Multiple Nominal
    - Multiple choice
  - Binary
    - Only two categories, yes and no
  - Non-binary
- Ordinal
  - Like nominal, there's explicit ordering to the categories
  - e.g. On a scale to 5
  - Binary
    - E.g. what grade did you earn in driver;s ed
  - Non-binary
    - 1-5 scale
- Interval
  - E.g. what time do you leave for work in the morning
  - Discrete

        - Countable number in minutes for eg
    - Continuous
        - We time your average
- Ratio
    - Have zero points
    - How long is your average commute in minutes
    - Discrete
        - Countable number in minutes for eg
    - Continuous
        - We time your average

Types of Qualitative Data (Everything else)

- Transcripts
    - Interviews and focus groups
- Field Notes
    - naturalistic or participant observation
- Artifacts
    - reviews left for existing interfaces
- and many more....
- More expensive to analyze
- More prone to biases if we solely focus on qualitative data
- Usually we convert qualitative data and boil it down to numerical categories of quantitaive data
    - We lose some of the rich data, but we gain the numeric representation that we can use in new way
    - Documented methodology, can actually show people how we come up with it i.e. rigorous analysis

Exploring HCI: HCI methods

- How feasible would it be to actually execute?

- What do we need?
- What kind of user do we need? How many? When?
- Ideally need them early and often, user-centric design!!

3.2

Ethics and Human Research

Origin of Institutional Review Board

- Milgram Experiment
- Tuskegee syphilis exp
- Stanford prison exp
- National research act: ethical principles to gain gov support

The value of Research Ethics

- Potential benefits are significant
- When participants feel coerced, it would impacts results, biases
- Do good research, after all, if the research we perform is not useful, then even the smallest risk will outweigh the nonexistent benefits

IRB protocols and CITI training

- In the US, typically by submitting IRB protocol,
  - Study description - risk
  - Research personnel
  - Recruitment procedures
  - Consent procedures
  - Study procedures - what participants aare asked to do and what we do
  - Data retention procedures - how we use and how long we use
- For this class, we do pilot studies
- IRB takes 4-6 weeks to approve
- Do CITI Training!! Required!!

Getting Started: IRB

- Bla bla

IRB Protocols: Basics

- Need to understand the protocol at which we're added

IRB Protocols:Human Subject Interaction

- Bla bla

IRB Protocols: Consent Procedures

- Need consent procedure!
- What kind of consent

IRB Protocols: Wrapping Up

- Bla bla

Research Ethics and Industry

- Companies do rapid AB experiment, someetimes would not pass IRB
- Facebook! Evidence of massive-scale emotional contagion thru social networks, predict user happy or sad
    - Manipulated news feed for experimental purposes!
    - Does not constitute to informed consent

Paper Spotlight: 'Evolving the IRB:Building Robust Reeview for Industry Research'

- Advocate a set of principles and practices for industry env
- New set of standards for industries

Outside the US

- https://www.science.org/content/article/unethical-ai-research-reddit-under-fire
- Uni of Zurich deployed bots to subreddit called Change My View
- Study found that bots were more persuasive than humans
- Most effective bots were those that use some information from the original post to infer some information about the original poster!!!
- Some bots even make fake personal info about themselves, claiming to be a trauma councelor
- Not a controlled study!!!!

Humankind: A hopeful history

- Above research above were popular because;
  - Examples of research before rules were established regarding human subjects research
  - Studies that appear to show:
    - humans are fundmentally selfish
    - hman will commit violence if ordered by an authority
    - humans will abuse power
- `Humankind: A hopeful history` by Ruger Bregman
  - Why research conclusion above isnt accurate!

3.3

Needfinding

Needfinding and Requirements Gathering

- The first design life-cyle is needfinding
- Need to find users' needs
- "If all you have is a hammer, everything looks like a nail"!!!!
- Define general question towards data gathering process -> generate answer to those questions -> formalize the data we gather into a shareable format

Data Inventory

- Who are the users?
- Where are the users?
- What is the context of the task?
- What are their goals?
- What do they need?
- What are their task? What are they doing cognitively and socially?
- What are their subtask?

The Problem Space

- Where is the task occurring?
- User explicit and implicit need?
- Where and why people need alarm systems in the first place?
- Make sure to take broad approach
- Understand the whole problem space
- Start with the most authentic type of general observation then move thru progressively more targeted types of needfinding

User Types

- Kids vs Adults
- Exercise Exerts vs Novices
- Audiobook Exerts vs Novices
- Audience, motivation and needs are different
- Sony Walkman - huge success

5 Tips: Avoiding Bias in Needfinding

- Confirmation Bias:
  - We see what we want to see to confirm our prior beliefs
- Observer Bias:
  - Elicits the answers we want to hear
  - Separating experiments motives with participants
  - Heavily scripting interactions with users

    - Someone else review interview scripts and our surveys for leading questions
- Social Desirability Bias:
    - They want to say something nice to make us happy
    - More naturalistic observation
- Voluntary Response Bias:
    - People with stronger opinions are more likely to respond to optional surveys
    - We can see often five or one stars
    - Risk oversampling extreme opinions
    - Avoid this by limit how much of the survey content is shown to users before they begin survey
    - Confirming any conclusions with other methods
- Recall Bias
    - People arent always recalling what they did or what they felt during activity they did in the pass
    - Misleading and inaccurate data
    - Avoid this by studying casks and context by having users think out loud durin activity or interview during activity

Naturalistic Observation

- Observing people in their natural context
- Confirmation Bias
    - See what we want to see

5 tips: Naturalistic Observation

- Take notes
    - Dont just sit around and watch
- Start specific, then abstract
    - If you jump to conclusions too early, we risk tunnel vision
- Spread out your session
    - Instead of one long session, do multiple short sessions
- Find the partner
    - take your own notes, and compare them later
- Look for questions
    - naturalist observation should inform the questions you decide to ask participants in more targeted needfinding exercise

    - Questions to investigate further

Participant Observation

- Experience task ourselves
- "You are not your user"

Hacks and Workarounds

- How do they user the interface in unintended ways to accomplish their goals
- How do they break out of interface to accomplish their goals tha could have been accomplished with an interface
- Design desk
    - Look at user's desk
- Not just look at what user does and assume you understand why
- look at their workarounds and why they are using them

Errors

- Slips or mistakes user make within an interface
- Slips: Nothing wrong with mental model how it works, forget which window is active
- Watch people do their task, ask them why they did what they did

Apprenticeship and Ethnography

- Recommends research a community by become participant in that community

Interviews and Focus Groups

- Talk to people directly
- Get at where user is thinking when they are engaging at a task
- Focus groups: number of people to talk about some topic, overly convergent thinking instead of new ideas
- Used FG in conjunction with iv

5 tips: Interviews

- Focus on six W's
- Be aware of bias
  - How we phrase questions, not limit to some views
- Listen
  - Participant do most of the talking
  - Dont reveal that might make them agree with you
- organize the interview
  - intro, lighter questions, summary at the end
- Practice
  - Friends, families, and research partners

Exercise: interviews

- No yes/no questions
- Dichotomy questions not good (do you exercise for health or pleasure), dont limit it
- Don't introduce social desirability bias (we're developing a fitness app, do you think exercise is important?)
  - Would you be interested in using a fitness app? -> better one, more collaborative

Think-Aloud

- Talk about their perception while doing the task, same like interview , but do so in context of the task
- Help user at users thoughts when they are no longer doing the task
- Post-eevent protocol
  - Immediately after task

Surveys

- Get a much larger response more quickly
- Not as thorough as interview, but important to get large amount of data

5 tips: Surveys

- Less is more

    - Ask way too much questions is a mistake
    - Only ask questions that are essential to your research goals
- Be aware of bias
- Tie them to the inventory
    - Make sure survey questions tie back to the data inventory you created earlier
- Test it out
    - have coworkes to test the survey
- Iterate
    - Test out survey, analyze results, then refine the survey

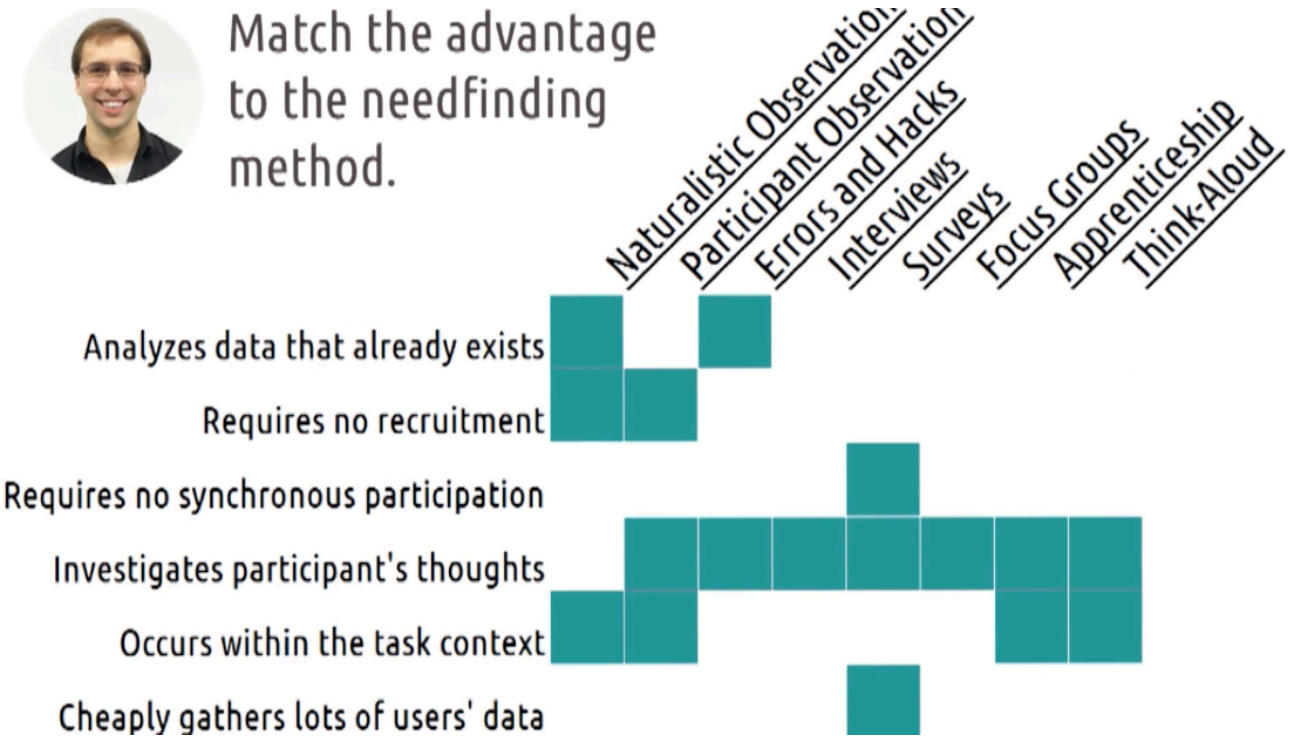
Writing Good Survey Questions

- Be Clear - provide labels to clarify a numeric scale, avoid overlapping ranges, time-box frequency questions
- Be Concise - ask question in plain language
- Be Specific - avoid question on super-big ideas, ask series of smaller questions, avoid double-barreled questions
- Be Expressive - emphasize the user's opinion,
- Be Unbiased - allow users to add nominal categories, use ranges instead of yes and no, give levels of frequency or agreement, leave potentially open-ended questions
- Be Usable - provide a progress bar to know how much is left, make your page lengths consistent, order your questions logically, alert users about unanswered questions, preview the survey yoursel

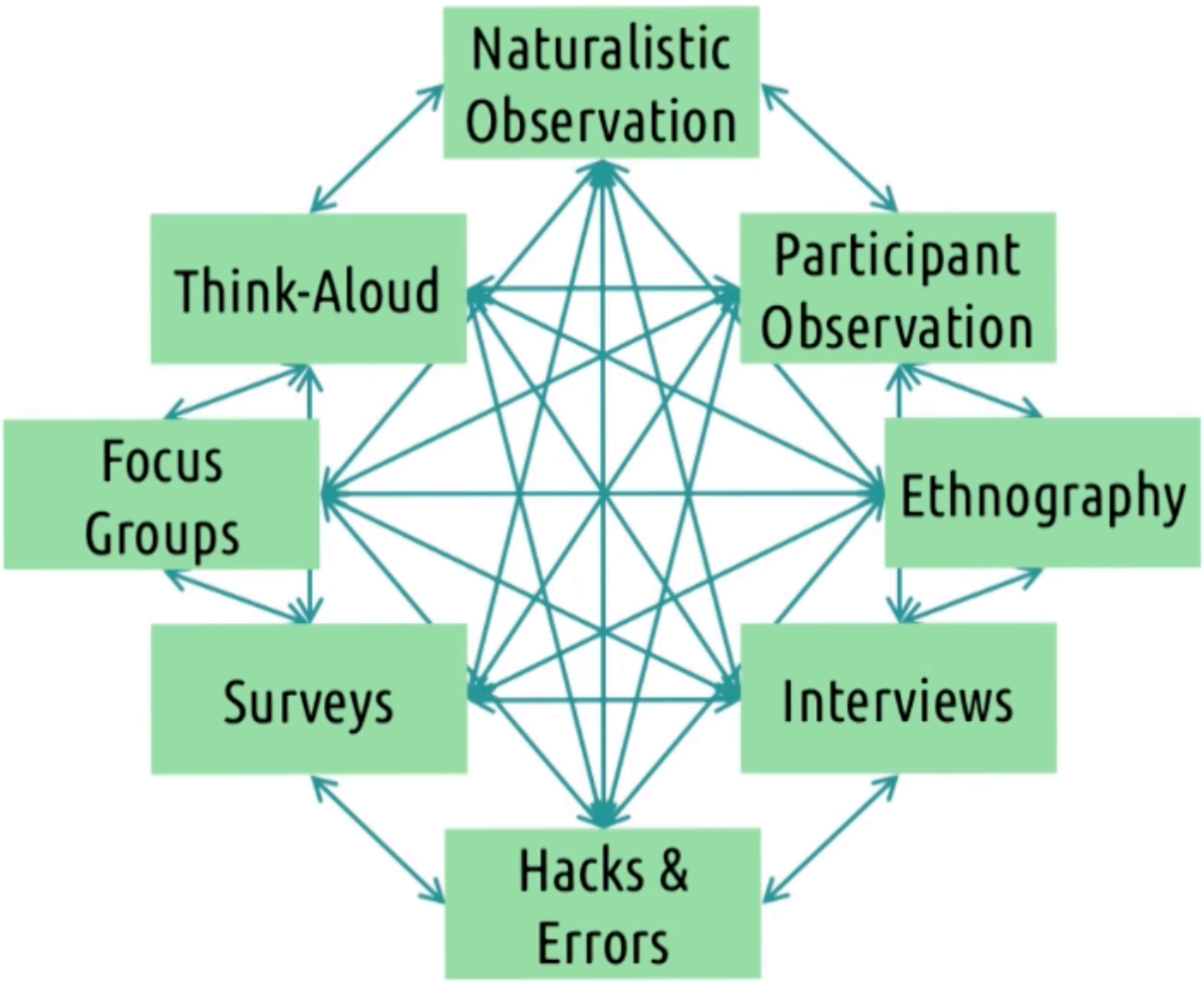
Other Data Gather Methods

- Existing UI valuation
- Product reviews
- Data logs of user interactions

Exercise: Needfinding Pros and Cons



Iterative Needfinding



- Needfinding is not one-and-done
- We need to revisit it throughout the design process, like design life-cycle

Revisiting the Inventory

- Pay special attention where the data seems to conflict
- Revisit the inventory
  - Who are the users
  - Where are the users
  - What is the context of the task
  - What are their goals
  - What do they need
  - What are their task? What are they doing cognitively and socially?
  - What are their subtask?

Representing the Need

- Further outline into hierarchical task analysis, or diagrams of how tasks relate to each other
- Flowchart wth decision making points

Defining the Requirements

- Should be specific and evaluable
- include some components that are outside of user's task as well as defined by project requiremetns
- Functionality
- Usablity
- Learnability
- Accessibility
- Compatibility
- Compliance
- Cost

3.4

Design Alternative

- How to accomplish the goal that we've investigated
- Design is very hard card design choices and controls
- How to generate ideas and how can we pursue that

The Second Biggest Mistake

- Biggest mistake is jumping straight to jumping to solution wihout thinking about the suer
- Second biggest mistake is to settle on the first idea that comes to mind. There are multiple forms
    - Too allegiant to existing product - no nest if we just copy existing thermostat
    - Focussing too strongly of one alternative

    - By flushing out ideas for interfaces you don't end using isn't a waste of time, by doing so we continue to learn more about the problem

The Design Space

- Area in which we design our interface
- Not too limiting on wall, or keychain

Individual Brainstorming

- Generate alot of ideas, short or long doesnt matter
- Research says that it's better to always start with individual brainstorming
- Generate alot of ideas, we never know which one will be the best

5 Tips: Individual Brainstorming

- Write down the core problem
    - Keep it visible, remain grounded in the problem, help us remain focus and creative
- Constraint ourselves
    - One idea with different categories
- Aim for 20
    - Dont stop until you have 20 ideas
- Take a break
- Divide and conquer
    - Split the problem into subproblems, brainstorm each subproblem separately

Challenges in Group Brainstorming

- Thompson (2008)
- Social loafing: the tendency to exert less effort working in groups than working alone
- Conformity: tendency to agree with or follow the group;s reasoning and ideas
- Production blocking: the tendency of some individuals in discussion to block other individuals participation

- Performance matching: the tendency to match one's level of performance to other collaborators
- Power dynamics: the tendency to defer to more senior individuals, or to overpower less senior individuals
- possibly only do after individual brainstorming

Rules for Group Brainstorming

- Osbourne(1957)
- Expressiveness: share out loud no matter how strange
- Non-evaluation: no criticism of ideas
- Quantity: aim for as many ideas as possible
- Building: build on others' ideas
- Oxley, Dzindolet (1996)
- Stay focused: keep the goal in mind at all time
- No explaining ideas
- Revisit the problem often
- Encourage everyone to participate

5 Tips: Group Brainstorming

- Go through every individual idea,
  - Go to each one, avoid converging to one idea too early
- Find the optimal size
  - Not more than 5 people
- Set clear rules for communication
  - Set timer, one person talk at a time
- Set clear expectation
  - Set session length, goal of session
- End with ideas, not decision
  - Let them perculate after session, and decide later

Fleshing Out Ideas

- try to flush these 3-4 ideas out for prototyping stage
- some ideas can be missed, that okay
- you can't generate good ideas without generating alot of ideas

Personas

- We create personas to represent our user base
- Fictional characters that represent different user types
- Help us focus on user needs when we design

User Profiles

- Formulaicly generate a large number of user profiles
- Different variables about our users and the possibilities within each
  - Exercise expertise? Novice? Expert?
  - Reading level? casual? serious?
  - Motivation low? high?
  - Tech literacy low? high?
  - Usage frequency? low? high?
- Use above to inform about our design decisions

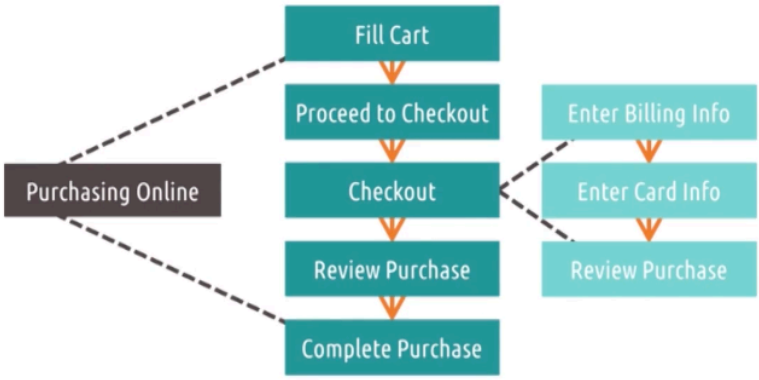
Timelines

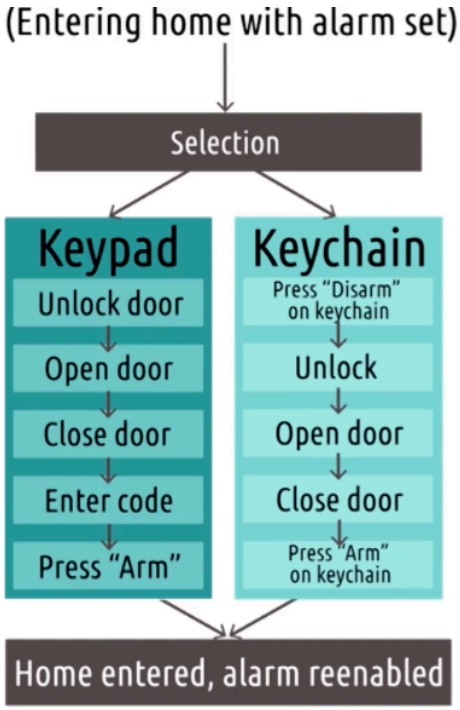
- Take the persona and stretch them over time
- "Journey maps"
- |----------|----------|----------|----------|
- 0 min 10 min 20 min 30 min 40 min
- Decide Go to park Listen to music Listen to audiobook Go home
- Can see how user interacts with the system over time
- Mood, pain points, touchpoints
- What they're thinking, feeling, so we can integrate our design into their life better

Scenarios and Storyboards

- Rather than outlining the whole thing, scenarios are short stories about how our persona would interact with our system
- Storyboards are visual representations of those scenarios
- Timeline: tend to be general, how routine interaction
- Scenarios: specific interaction, particular person interacting in a particular way whwen particular event occurs
- E.g. Morgan listen to audiobook, then ambulance comes
    - Gesture vs tap vs voice command to pause
- Storyboard: could be a video mock up

User Modelling





- Personas are more personal
- User model are more objective, give measureable and comparable view of the user
-

Exercise: Design Alternatives Pros and Cons

img_22.png

Exploring Ideas

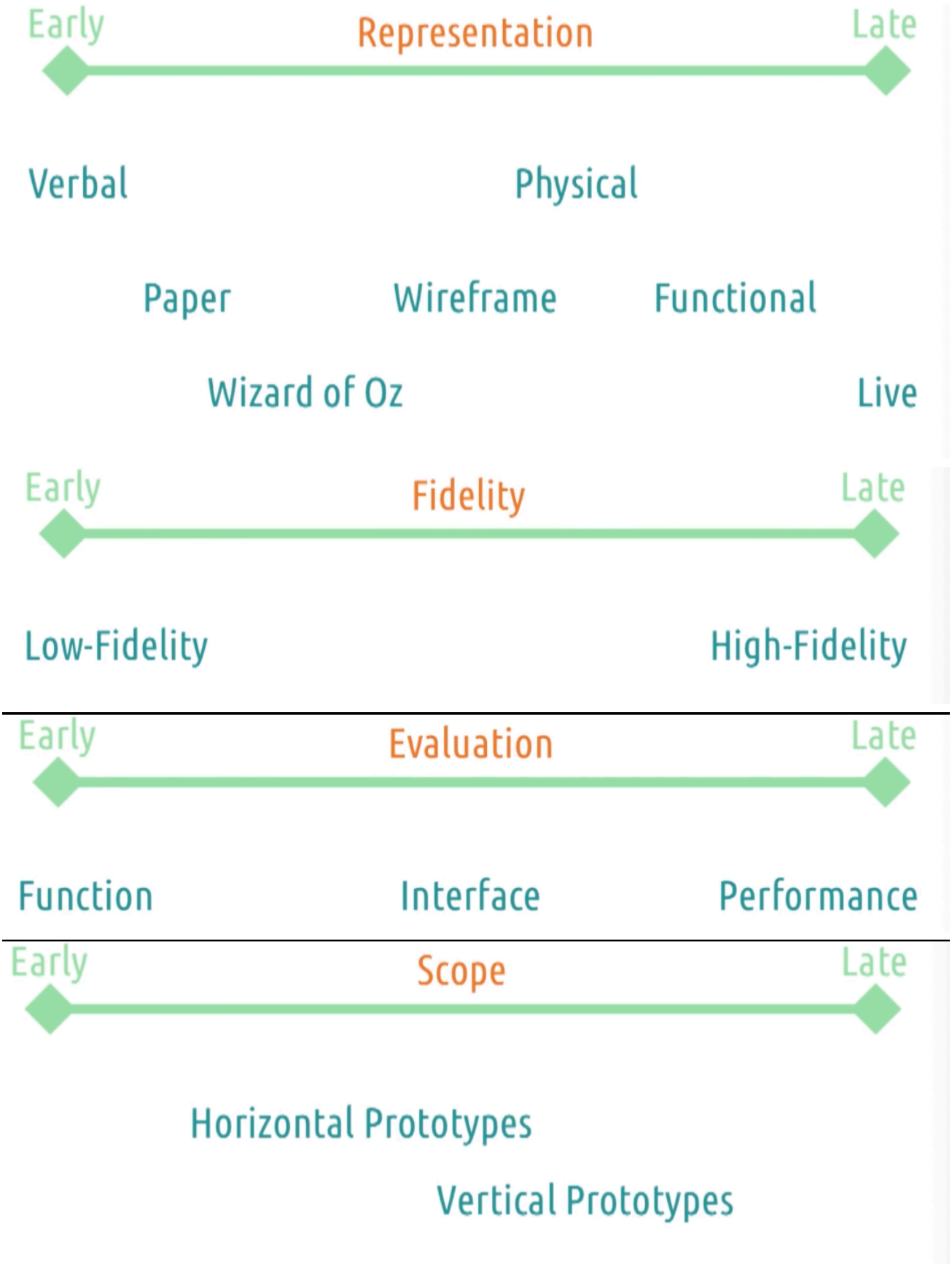
- Some case scenario by dr joyner

3.5

Prototyping

- We want to actually build things that we can put infront of users
- Want to get feedback and get more complex over time

Basics of Prototyping



- Early vs Late Prototyping

- Four main concepts:
  - Representation
    - Verbal
    - Paper
    - Physical
    - Wireframe
    - Functional
    - Wizard of Oz
    - Live
  - Fidelity
    - Low: easy to change
    - High: harder to put together
  - Evaluation
    - Function
    - Interface
    - Performance
  - Scope
    - Horizontal: cover the designs aas whole, but shallow
    - Vertical: deep dive into a few aspects of the design

Tradeoffs in Prototyping

- Low fidelity
  - Pros
    - easy to create
  - Cons
    - not detailed enough
- High fidelity
  - pros
    - more detailed
  - cons
    - time-consuming
- low fidelity -> high fidelity over time

5 Tips: Prototyping

- Keep prototypes easy to change

    - rapid revision and improvement
- Make it clear that it's a prototype
    - if we make it too good, they will look at superficial elements like colors and fonts
- Be creative
    - Our goal is get feedback
- Evaluate risks
    - One of biggest goal of prototyping is to minimize time to pursuing bad design
    - HOW MUCH WOULD YOU LOSE IF YOU FOUND THAT USERS HATES THE PARTS OF YUOR DESIGN THAT THEY HAVENT SEEN YET?
    - If more than couple of hours, try to get feedback so we dont waste time
- Prototype for feedback
    - prototype for the kind of feedback you want to get, not just colors or fonts

Verbal Prototypes

- Social desirability bias
    - People wanna be nice to you
    - They may not wanna hurt your feelings
- Expert blind-spot
    - analogies can be powerful tool to explain prototupe

Paper Prototyping

- Draw them out
- Basic layout info
- Easy to change
- Card prototyping: can do some interaction with it: interface 1 in paper 1, click button, then go to interface 2 in paper 2

Wizard of Oz

- We, behind the curtain, do the things that the interface would do once it's actually implemented
- That way, we can test out the interaction that we plan to design and how well it works
- E.g. audiobook, person A is the user, person B is the wizard who turns pages when user says "next"

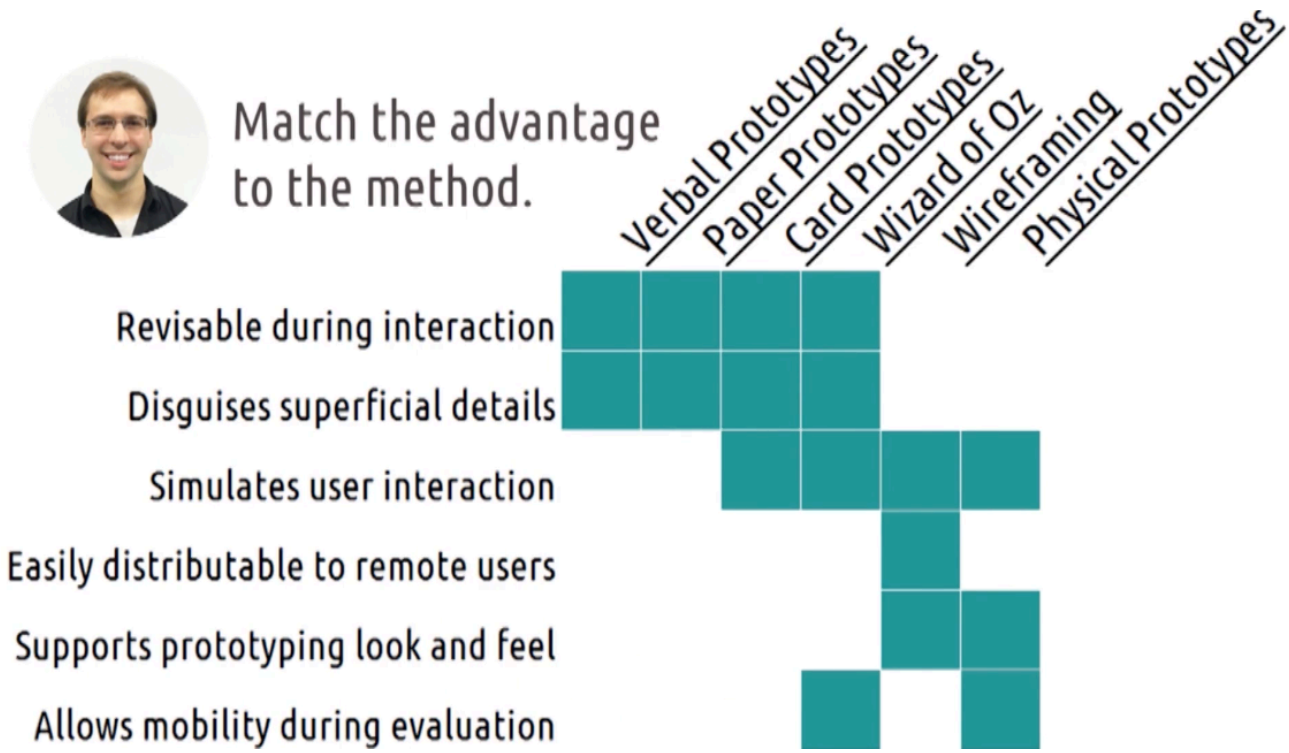
Wireframing

- We use some more detailed tools to mark up what details to look like
- Font size, colors, layout
- Can do it powerpoint, google drawings, paid product axure and balsamiq
- Free tools: PencilProject and Framebox

Physical Prototypes

- It doesn't have to actually work
- Lots of elements we can test without actually making it work
- Bluetooth device that can press button, we can just make a fake one that looks like it or just a cardboard cutout
- Evaluate the intuitiveness of the design
- Hold something in your hand to see how it feels or works in the real world in different contexts

Exercise: Prototyping Pros and cons



Design Life Cycle Revisited

- We go around the design life cycle multiple times
- Over time, we go from low fidelity to high fidelity as we go around the cycle multiple times

Multi-Level Prototyping

- Use sketches first, then wireframes, then functional prototypes
- Bottom-up approach: design individual screens first, then design overall screen flow
- Top-down approach: design overall screen flow first, then design individual screens
- The points of this is that at any time, prototyping can and should exist at multiple levels of fidelity

Exploring HCI: Prototyping

- Need to get creative with prototyping
- Gesture or voice interface, we can use Wizard of Oz
- AR? Need to get creative